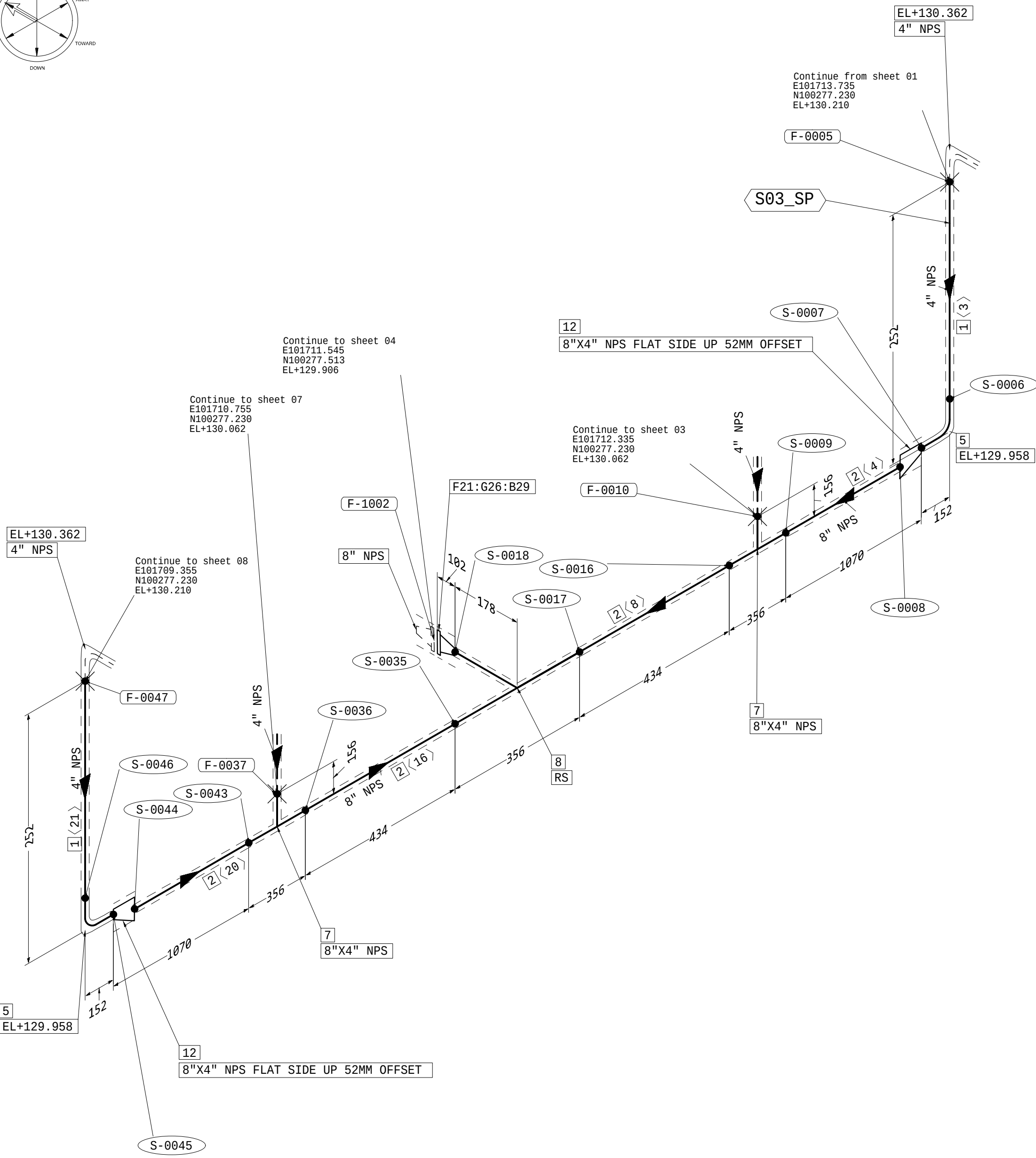
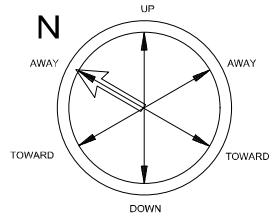


THIS SPOOL MARKUP DRAWING IS GENERATED FROM ENGINEERING PCF & REFERRING APPROVED ENGG ISOMETRICS. ENGINEERING ISOMETRICS TO BE REFERRED FOR REVISION INFORMATION & IN CASE OF ANY DISCREPANCY DATA GIVEN IN ENGINEERING ISOMETRIC SHALL GOVERN



MATERIALS				STANDARD DETAIL REFERENCE DRAWINGS																			
PT No.	DESCRIPTION	NPD(IN)	ITEM CODE	QTY	DETAILS OF HEAVY WELDING BOSS FOR THREADED CONNECTIONS CONSTRUCTION DETAILS AND DETAILS FOR ISOMETRIC PIPING INSULATION DETAILS FOR HOT PIPING INSULATION DETAILS FOR COLD PIPING INSTALLATION OF JACK SCREWS FOR FLANGED JOINTS																		
PIPES					AE-036175 AD-036077 AA-036911 AC-036933																		
1	PIPE SMLS 4" A106-B SOUR STD BE B36.10M	4	PPS2AERT04ZZABUAIA	0.200	NOTES 1. MATERIAL SHOWN WITH A BROKEN LINE AND SPOOLS LISTED IN THE SAUDI ARAMCO FABRICATION LIST ARE NOT TO BE FURNISHED BY THE VENDOR. 2. PIPING DETAILS SHALL BE CUT AT ALL POINTS MARKED F.W. (FIELD WELD). 3. EACH FABRICATED SECTION SHALL BE CLEARLY MARKED WITH ITS CODE NUMBER. 4. UNLESS OTHERWISE NOTED: (A) ALL DIMENSIONS ARE TO FACE OF FLANGES. (B) BOLT HOLES ON FLANGES STRADDLE C. (C) WELDING ELBOWS ARE LONG RADIUS. 5. FABRICATION SHALL BE IN ACCORDANCE WITH SAUDI ARAMCO SPECIFICATION 01-SAMSS-10. 6. GASKET ALLOWANCE 7. PIPING MATERIAL IS <table border="1"><thead><tr><th>PIPE SIZE</th><th>SCHEDULE</th><th>SPECIFICATION</th></tr></thead><tbody><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr></tbody></table> 8. WELDING STRESS RELIEF - YES ☐ NO ☐ SPOT X-RAY 01-SAMSS-10 ☐ 9. TESTING SHOP HYDROSTATIC TEST - YES ☐ NO ☐ THE SHOP HYDROSTATIC TEST SHALL BE CARRIED OUT ON PIECES HAVING ALL ENDS FLANGED. HYDROSTATIC TEST PRESSURE: 10. ALL PIPING MATERIAL IS IN ACCORDANCE WITH SAUDI ARAMCO ENGINEERING SPECIFICATION SAESL-105 CLASS ☐ ALL EXCEPTIONS ARE NOTED ON THE ISOMETRIC. 11. INSULATION REQUIRED - YES ☐ NO ☐ INSULATION SHOWN ON DRAWING				PIPE SIZE	SCHEDULE	SPECIFICATION												
PIPE SIZE	SCHEDULE	SPECIFICATION																					
2	PIPE SMLS 8" A106-B SOUR STD BE B36.10M	8	PPS2AERT08ZZABUAIA	2.704																			
FITTINGS																							
5	ELB 90° SMLS LR 4" A234-WPB SOUR STD BE B16.9	4	PE92AZRT04ZZABUIQ	2																			
7	TEE RED. SMLS 8" X 4" A234-WPB SOUR STD BE B16.9	8X4	PTS2AZRT0804ABUIQ	2																			
8	TEE EQ. SMLS 8" A234-WPB SOUR STD BE B16.9	8X8	PTF2AZRT08ZZABUIQ	1																			
12	RED. ECC. SMLS 8" X 4" A234-WPB SOUR STD BE B16.9	8X4	PRH2AZRT0804ABUIQ	2																			
FLANGES																							
21	FLG WNRF 8" A105N SOUR STD, 150# 125 B16.5	8	PFY2ADRT08ZZACPADH	1																			
GASKET																							
26	GSK SPR WND W/CS CENTERING RING + I - SS 8" 4.50 MM THICK SS316L+GRPH FILL SOUR 150# B16.20	8	PGUUB5RT08AIAAAZZC	1																			
FASTSNERS																							
29	BOLT, STUD+2 NUTS A193 GR B7M/ A194 GR 2HM, CERAMIC FLP COAT SOUR B18.2.1/18.2.2 - 120 mm length	0.75	PH8ASCRTAG00120ZZF	8																			

PIPE CUT LENGTH SUMMARY							
SPL No.	NPD	PC No.	LENGTH	ALLOWANCE	TOTAL LENGTH	ISO For	
S03_SP	8	<8>	0.434	0.000	0.434	SXX_Spool Sequence Number (S01,02,03,04,05,06,07,08,09,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100)	
S03_SP	8	<4>	0.918	0.000	0.918	SXX_SP - Fab. Spool	
S03_SP	4	<3>	0.100	0.000	0.100	SXX_SP - Fab. P to P Doubling (0.0M+0.0M)	
S03_SP	8	<16>	0.434	0.000	0.434	SXX_SP - Erec. Random Pipe (0M,12M)	
S03_SP	8	<20>	0.918	0.000	0.918	SXX_SP - Erec. Part Loose Fittings	
S03_SP	4	<21>	0.100	0.000	0.100	SXX_SP - Erec. Range (0F-Loose Flange, 0F5-Bolt, 0F5-Spec/0.0)	

NOTES :		BUTT WELD		FIELD WELD	
1. ALL COORDINATES AND ELEVATIONS ARE IN 'METER' AND ALL DIMENSIONS ARE IN 'MM'.					
2. PIPE QUANTITY IS IN METERS AND OTHER QUANTITY IN NUMBERS.					
3. FOR STANDARD SUPPORT, REFER DOC. NO. "PR-868-ST-0001".					
4. IN CASE OF ANY DISCREPANCY FOR LINE CONDITIONS, LINE DATA GIVEN IN LINE LIST SHALL GOVERN					
5. FOR SMALL BORE LINES, DIMENSION & ROUTING SHALL BE FIELD ADJUSTED.					
6. THE SPECIFIED THEORETICAL LOCATION OF EQUIPMENT NOZZLES, SUPPORTING STRUCTURE SHALL BE VERIFIED BEFORE INSTALLING THE PIPING & INTOLERABLE DISCREPANCIES SHALL BE DISCUSSED WITH DESIGN SITE ENGINEER					
7. ALL THREADED CONNECTION SHALL BE SEAL WELDED, EXCEPT DRAIN AND VENT PLUG.					
8. REFER LINE LIST.					
9. ALL PRESSURE WELDS, OTHER THAN BUTT WELDS, INCLUDING BRANCH CONNECTIONS SUCH AS BRANCH-OLETS, AND BOSS) SHALL BE 100% PT/MT EXAMINED. ALL ATTACHMENT WELDS ON PRESSURIZED COMPONENTS THAT ARE NOT HYDROTESTED AFTER MAKING THE ATTACHMENT WELD SHALL BE 100% PT/MT EXAMINED.					

NOTES CONT. :		PWHT		CHEMICAL CLEANING		PIPING MATERIAL CLASS		DESIGN		OPERATING		TEST PRESS.		TEST MEDIUM		RADIOGRAPHY %		INSULATION		TRACING		PAINTING		STRESS SYS NO.		LINE ID	
91. FOR SOUR SERVICE PWHT IS MANDATORY FOR 1. DOUBLE SIDED WELDING, 2. INTERNAL WELD REPAIR 3. INTERNAL WELD BUILD UP FOR DISSIMILAR THICKNESS JOINTS TO OVERCOME MISALIGNMENT.		Y	NOTE-8	1CH2P06	190	170	140	65	428	WATER	10	10	25	HC	NA	APCS-2A	867-GPT4-171	MDEA-17358-420A									

DESIGN CERTIFICATION		REVIEW FOR KEY DRAWINGS		OTHER	
ENGL. DATE	CERTIFIED BY DATE	CONF. ASH/KEY DATE	OPERATIONS DATE	BY DATE	
REV. NO.	DATE	REVISION DESCRIPTION			

JOB ORDER/REV NO.		THIS REVISION IS NOT TO BE USED FOR CONSTRUCTION UNLESS CERTIFIED AND DATED			
10-14815-0003					
SAUDI ARABIAN OIL COMPANY					
ISOM 8" - MDEA-17358-1CH2P06		DRAWING TYPE ISO			
MDEA SERVICE					
JFGP GAS PROCESSING TRAIN-4					
JAFURAH		SAUDI ARABIA			
PLANT NO.	INDEX	DRAWING NUMBER	SHT. NO.	REV. NO.	
867	L	YB-215336	02of08	00A	

SITE REV. NO.		RESTRICTED		SMART		CADD-03	
00A							